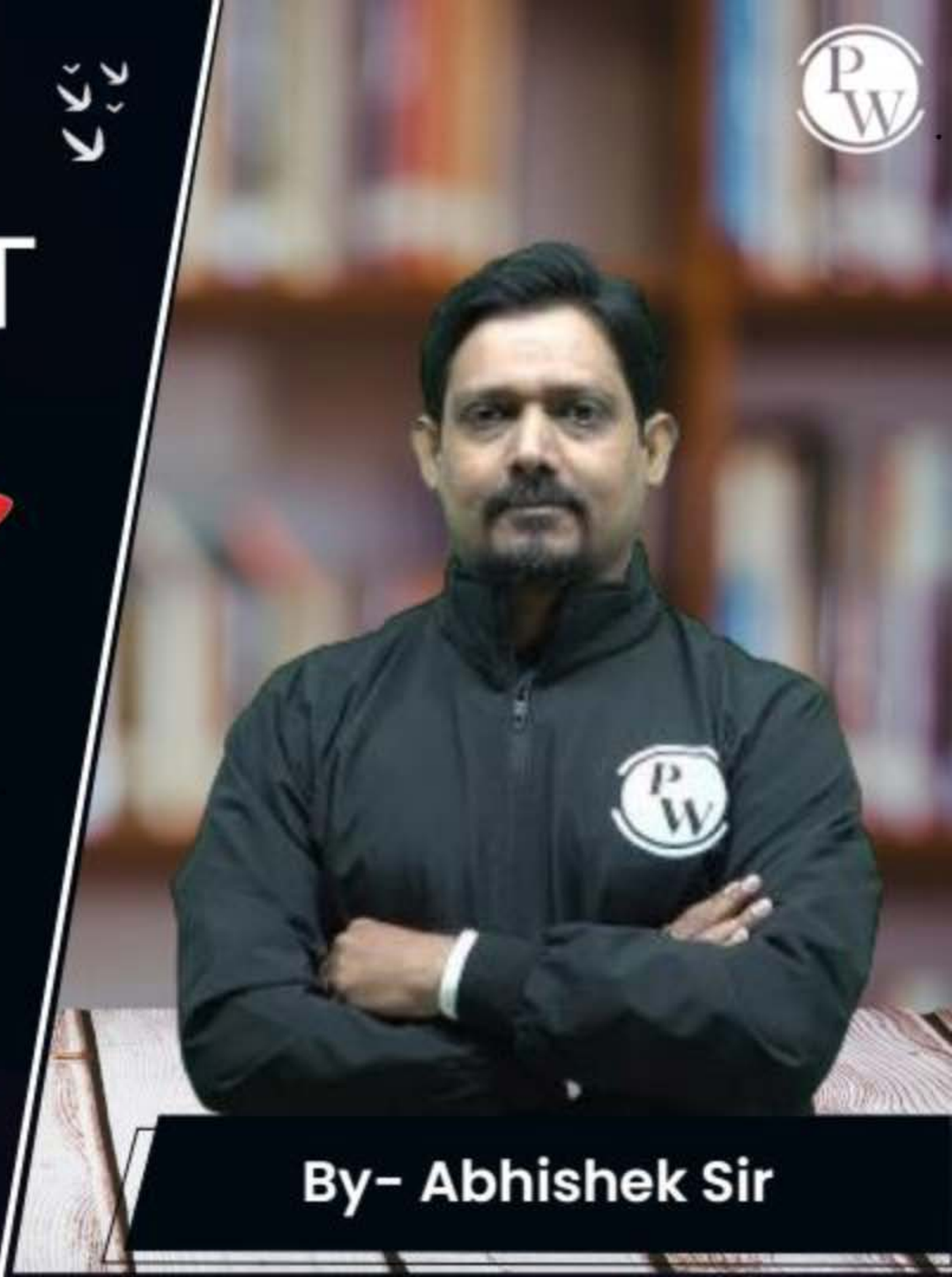
Computer Science & IT



C Programming

Functions & storage classes

Lecture No. 05



By- Abhishek Sir

Recap of Previous Lecture



Topic Recursion

Topic Recursion Tree

Topic

Topic

Topic

Slide

Topics to be Covered



- Topic — Practice problem
- Topic — Extern & Register
- Topic
- Topic
- Topic

Slide



Recursion

```
#include<stdio.h>
void print(int n)     {
      if (n <= 0)  return;
      print(--n);
      printf("%d", n);
      }

int main(){
      print(5);
      return 0;
}
```

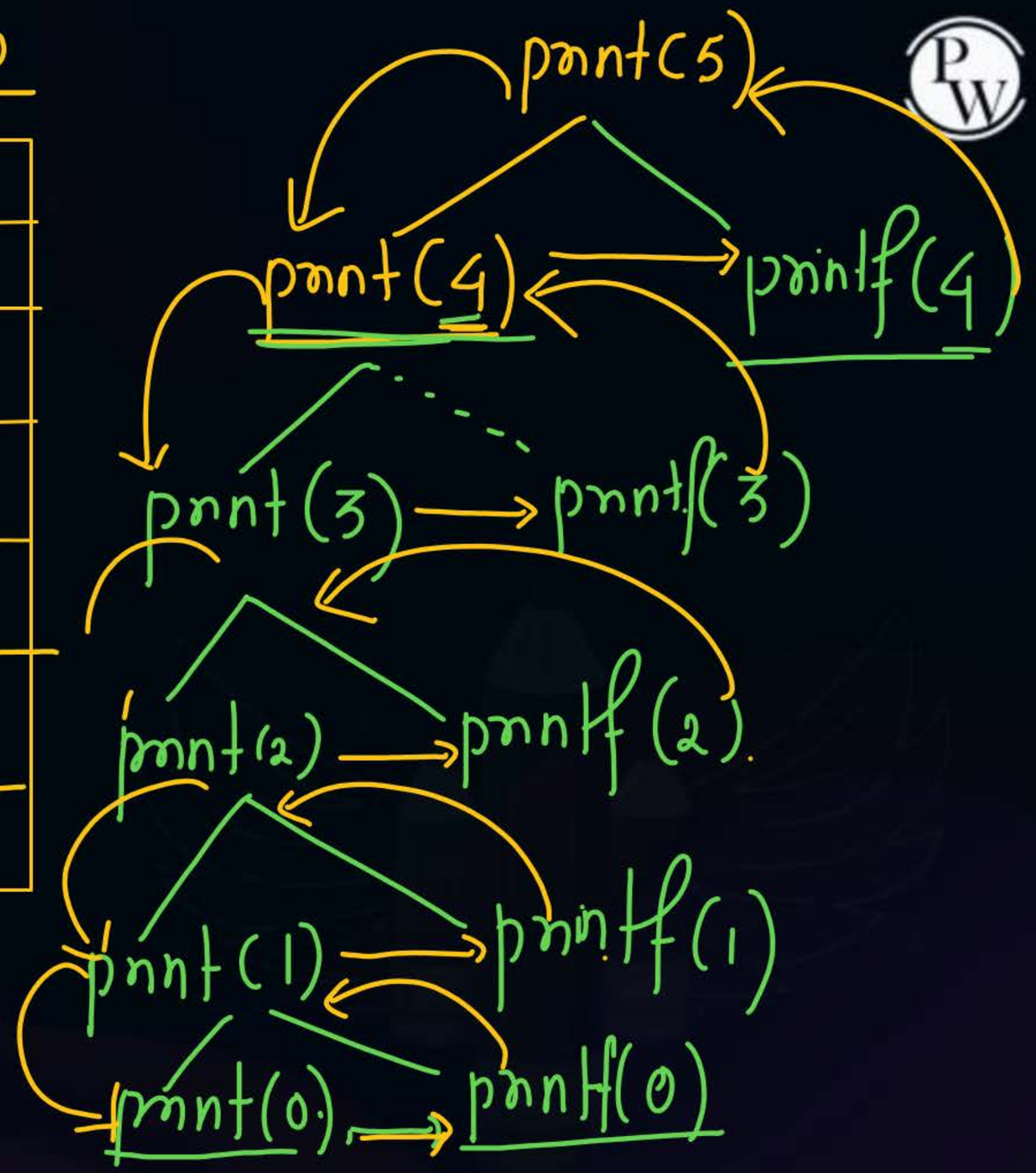
n is Local

mod

n = n - 1

auto





01234

Slide

tracing equals to control transfer.



Recursion

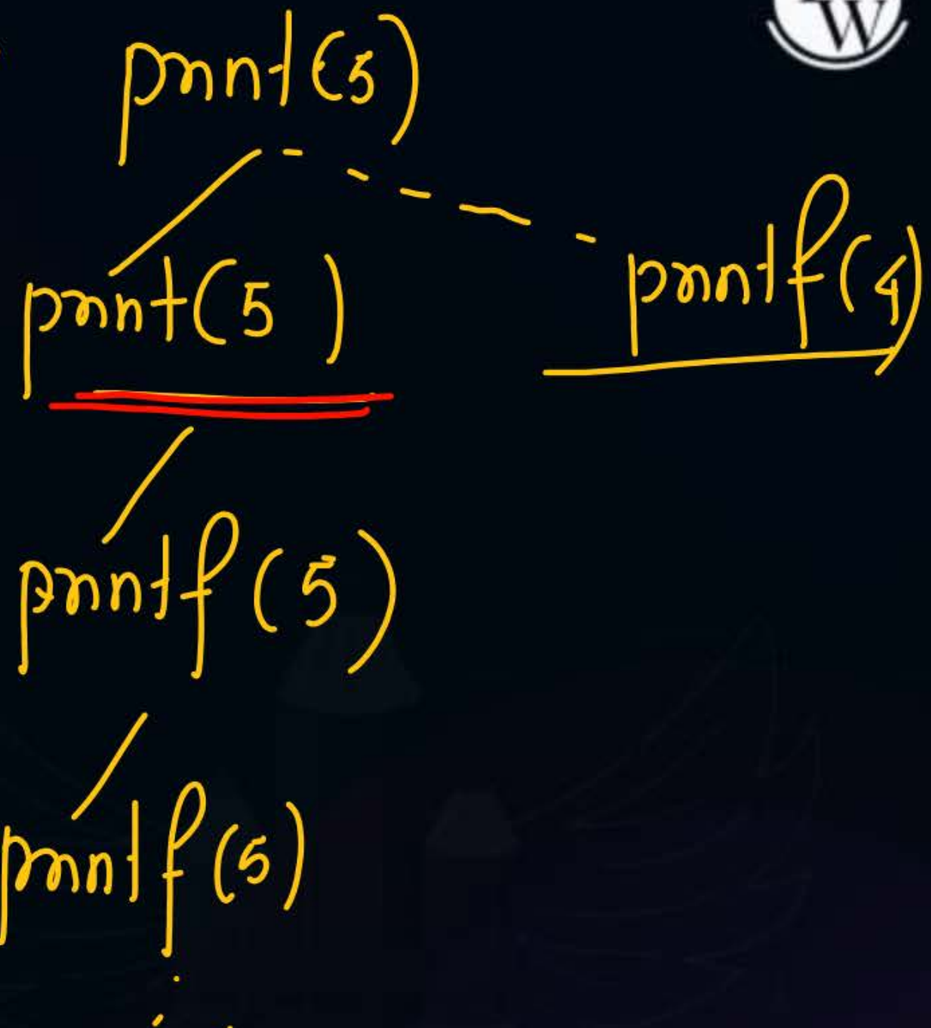
```
#include<stdio.h>
void print(int n)    {
     if (n <= 0)  return;
     print(n--);
     printf("%d", n);
     }

int main(){
     print(5);
     return 0;
}
```

post decrement



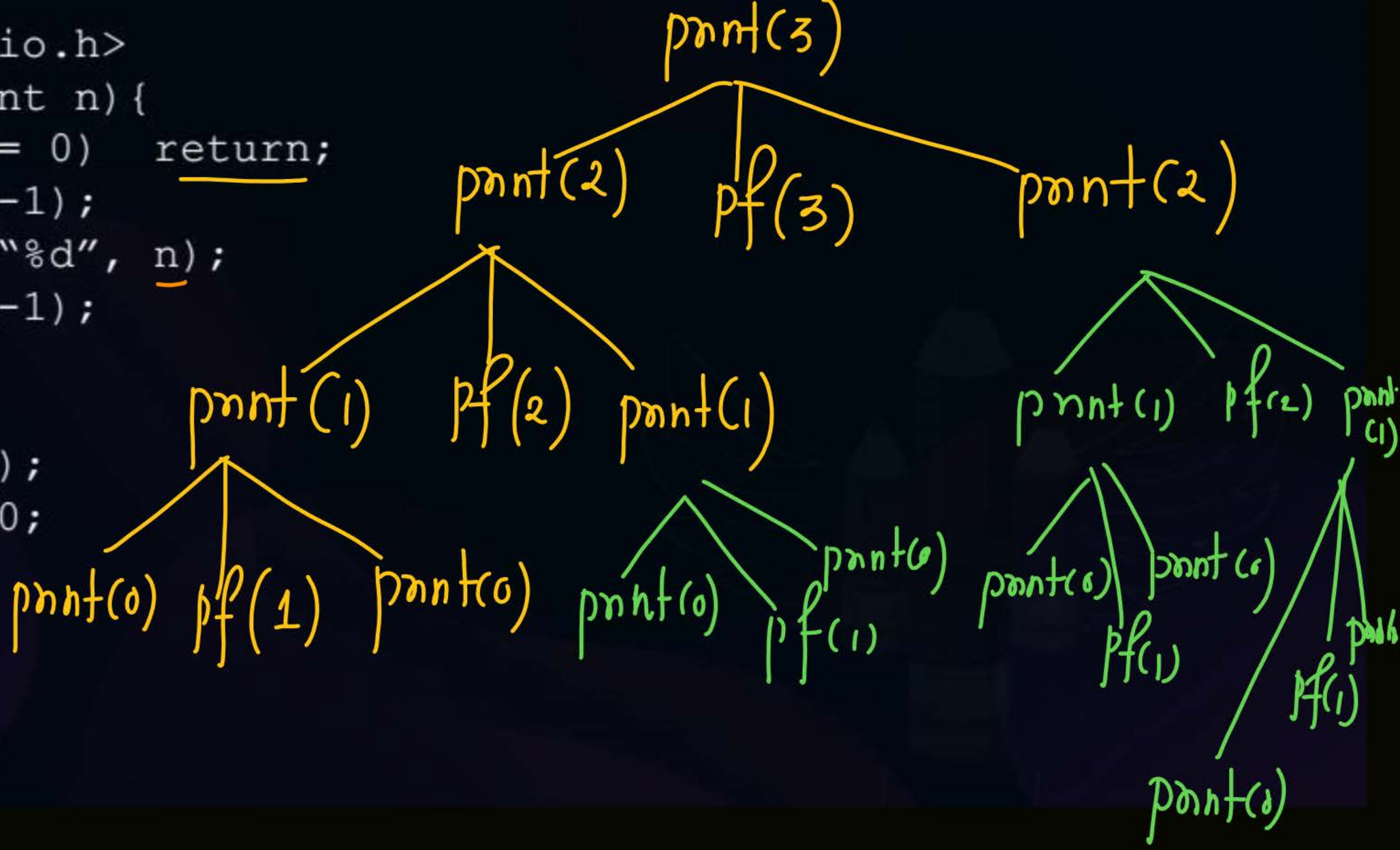




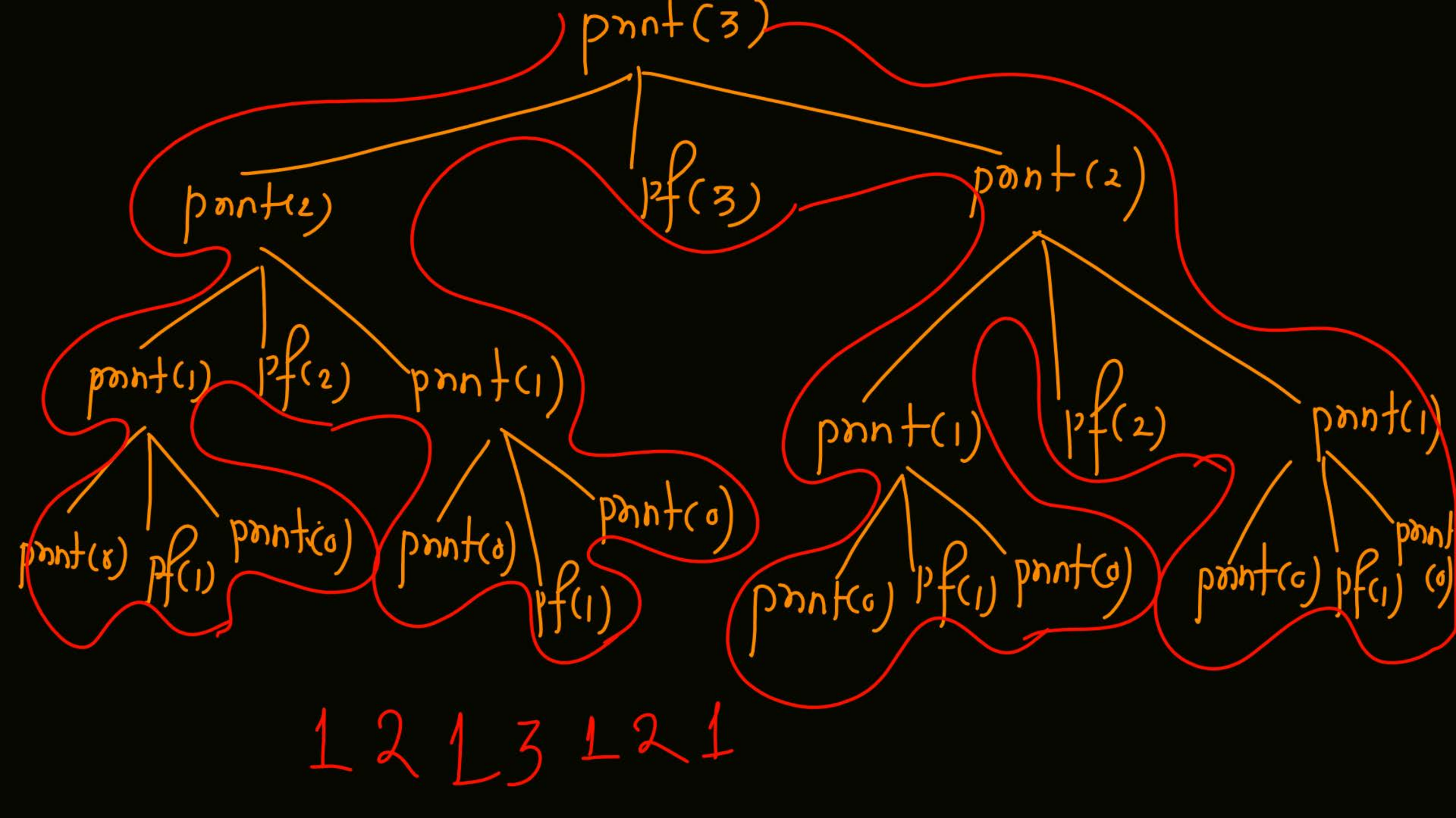
abnormal Termination

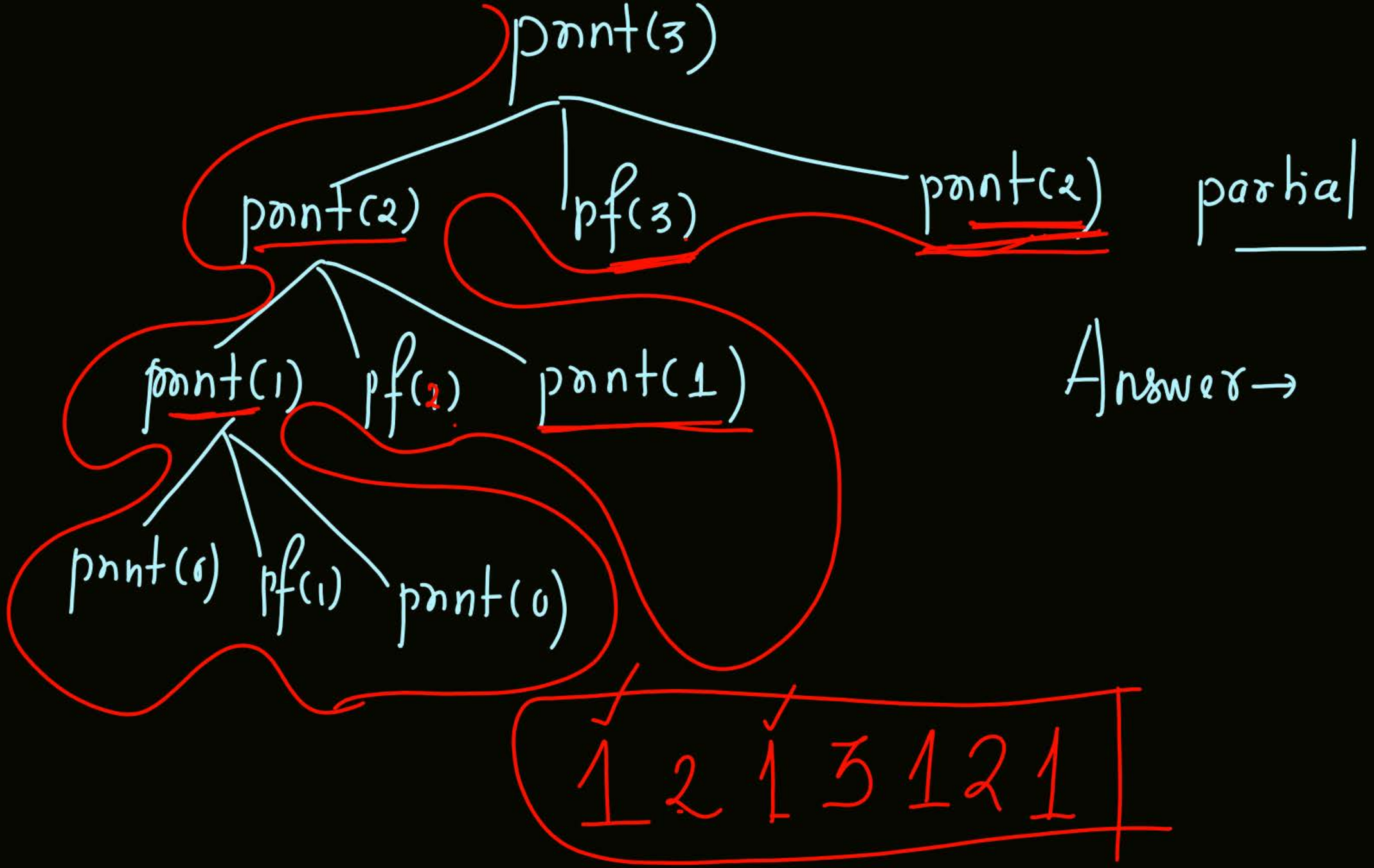
Slide

```c
#include<stdio.h>
void print(int n){
      if (n <= 0)   return;
      print(n-1);
      printf("%d", n);
      print(n-1);
}
int main(){
      print(3);
      return 0;
}
```



Slide


print(3)
print(2)
pf(3)
print(2)
print(1)
pf(2)
print(1)
print(1)
pf(2)
print(1)
print(0)
print(0)
pf(1)
print(0)
print(0)
pf(1)
print(0)
print(0)
pf(1)
print(0)
print(0)
pf(1)
1 2 1 3 1 2 1


print(3)
print(2)
pf(3)
print(2)
print(1)
pf(2)
print(1)
print(0)
pf(1)
print(0)
partial
Answer →
1 2 1 3 1 2 1



Recursion

Consider the following C code

```
int fun(int a)
{
    Static int i;
    i = 1;
    if (a ≥ 5)return a;
    a = a + i;
    i + +;
    return fun(a);
}
```

The value returned by fun(1) is _____

Static int i = 1; ← Initialization one time

assignment

i = 1;

i = i * 2,

fun(1) a = 1

i: 1 2 1 2 1 2 1 2 1

a = 2, return fun(2)

a = 2

a = 2 + 1 = 3

return fun(3)

a = 3 + 1 = 4

return fun(4)

a = 4 + 1 = 5

return fun(5)

Slide



Recursion

#Q. Consider the following program

```
#include<stdio.h>
int foo(int n){

        if (n<=9)
            return n;
        else
            return n%10+foo(n/10);
}

int main(){
        printf("%d", foo(12345));
        return 0;
}
```

A = 10
B. 11
C. 12
D. 15

Remainder

foo(12345) ← 15
return 5 + foo(1234) ← 10
return 4 + foo(123) ← 6
return 3 + foo(12) ← 3
return 2 + foo(1) ← 1 ; 1 <= 9
Base condition

1

Slide



Recursion

#Q. Consider the following program

```
#include<stdio.h>
int foo(int n){

      if (n<=9)
          return n;
    else
          return n%10+foo(n/10);
}

int main(){
      printf("%d", foo(12345));
      return 0;
}
```

A. 10

B. 11

C. 12

D. 15



Recursion

#Q. Consider the following program

```
int mystery(int n) {
        if (n <= 0)
                return 1;
        else
                return 3 + mystery(n - 1);
}
```

What is the ~~output the program~~ when mystery(3) is called

A. 9
B. 10
C. 11
D. 12

return value

mystery(3) 10
return 3 + mystery(2) 7
return 3 + mystery(1) 4
return 3 + mystery(0) 1

Slide



Recursion

#Q. Consider the following program

```
void Dosomething(int n) {
        if (n > 1)
                Dosomething(n-1);
        for (int i = 0; i < n; i++)
                printf("*");
        printf("\n");
}
```

The number of stars will print if the Dosomething(5) is called ?

7
10
16

15

DoSomething(5)

DS(4) for(i=0; i<5; i++)

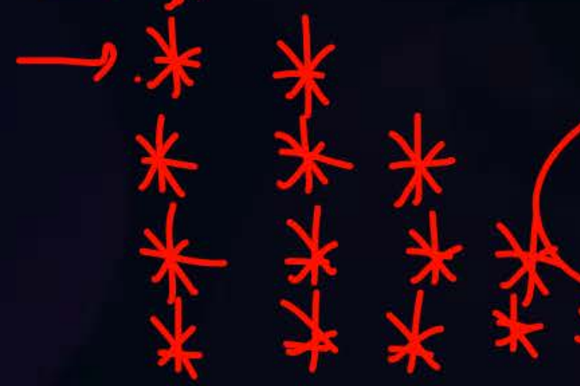
DS(3) for(i=0; i<4; i++)

DS(2) for(i=0; i<3; i++)

D(1) for(i=0; i<2; i++)

for(i=0; i<1; i++)

*****



Slide



Recursion

#Q. Consider the following program

```
void quiz(int i)
    {
        if (i > 1)
        {
            quiz(i / 2);
            quiz(i / 2);
        }
        printf("*");
    }
```

What va[redacted]

for quiz(5)

A. 8

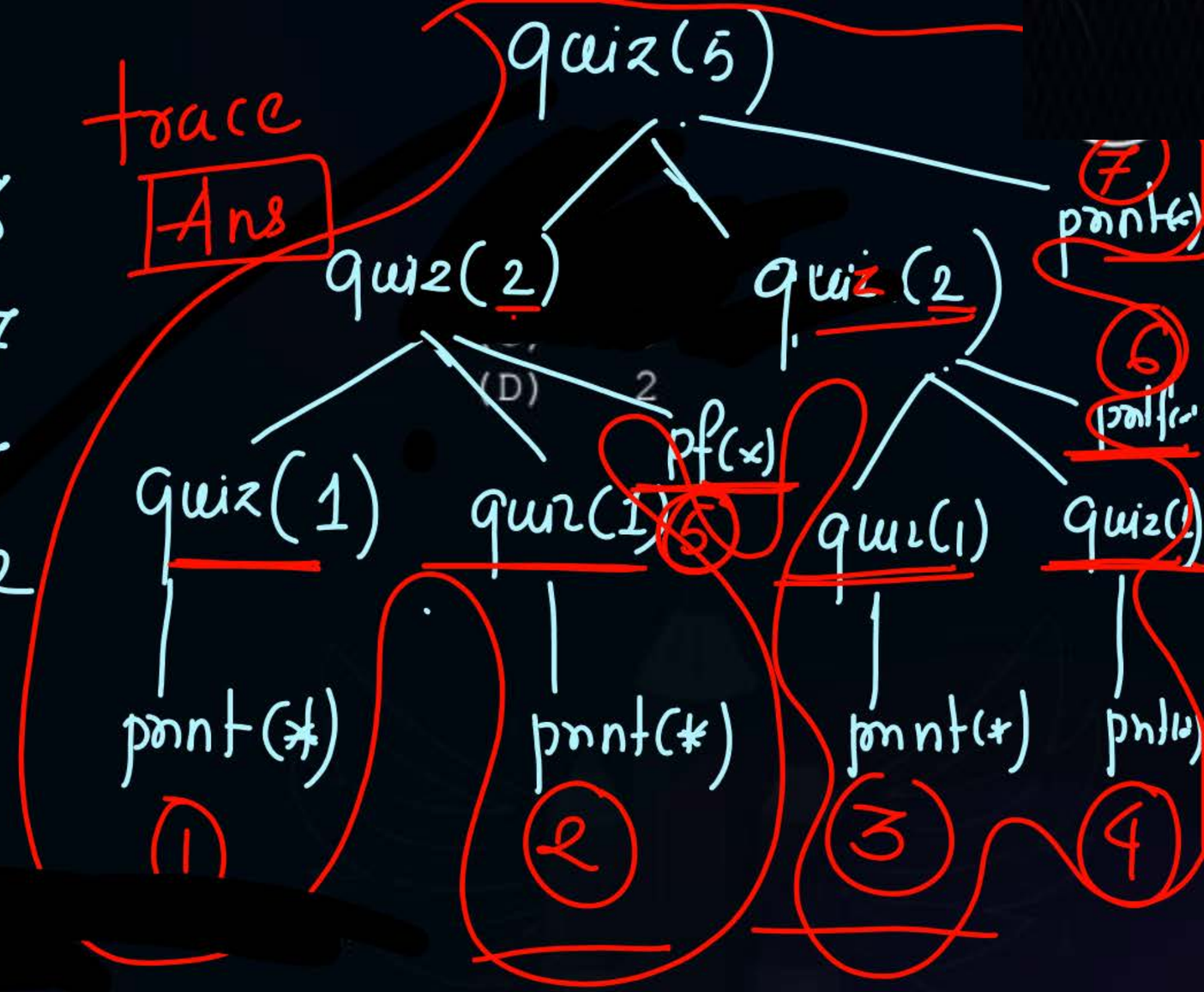
B. 7

C. [redacted]

D. 2

trace

Ans



Slide



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

Slide

THANK - YOU

